Tribhuwan University

Institute of Science and Technology

Bhairahawa Multiple Campus

An internship report on

"Frontend Development"

At Sajilo Digital Pvt. Ltd

(In partial fulfilment of the requirement of Bachelor of Science in Computer Science & Information Technology (BSc. CSIT))

Submitted By:

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Submitted To:

Department of Computer Science and Information Technology

Bhairahawa Multiple Campus

Bhairahawa, Nepal

November, 2025

Under the supervision of

Mr. Suraj Pandey

SUPERVISOR RECOMMENDATION

This is to certify that the internship report titled "Modernization and Redesign of Code for Change Nepal Website Using ReactJS" has been prepared by Bijay Kumar Chaudhary under my academic supervision. This report is submitted in partial fulfillment of the requirements for the Bachelor of Science in Computer Science and Information Technology degree.

As his college supervisor, I monitored Bijay's progress throughout his internship at Sajilo Digital Pvt. Ltd. He effectively applied his knowledge of modern web technologies to completely redesign and rebuild an existing website using ReactJS, demonstrating technical proficiency and understanding of component-based architecture. His report comprehensively documents the migration process, technical challenges, design decisions, and implementation details.

The report is well-structured, technically sound, and demonstrates a mature understanding of modern frontend development practices. It meets the academic standards expected for the BSc.CSIT program.

……

Mr. Suraj Pandey

Supervisor, BSc.CSIT

Bhairahawa Multiple Campus

LETTER OF APPROVAL

This is to certify that the internship report "Modernization and Redesign of Code for Change Nepal Website Using ReactJS" prepared by Bijay Kumar Chaudhary is submitted in partial fulfillment of the requirements for the Bachelor of Science in Computer Science and Information Technology degree. We have examined this report and find it satisfactory in terms of scope, content, quality, and organization. It adequately represents the work carried out during the internship period and fulfills the academic criteria for the degree

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ACKNOWLEDGEMENT

I extend my sincere gratitude to all those who supported and guided me throughout my internship journey and during the preparation of this report.

My deepest appreciation goes to Sajilo Digital Pvt. Ltd. for providing me with the invaluable opportunity to work as a Frontend Developer Intern. I am profoundly grateful to my company mentor, Mr. Shailesh Mishra, for his exceptional guidance, technical insights, and constant encouragement. His mentorship was instrumental in helping me successfully redesign and modernize the Code for Change Nepal website.

I extend my heartfelt thanks to my college supervisor, Mr. Suraj Pandey, for his academic guidance, constructive feedback, and support throughout the internship and report writing process. His supervision ensured my work maintained academic rigor while solving real-world problems.

I am also thankful to the entire technical team at Sajilo Digital for their collaborative spirit, knowledge sharing, and supportive work environment that greatly enriched my learning experience.

Finally, I express my eternal gratitude to my family and friends for their unwavering support, patience, and encouragement throughout my academic and professional journey.

……..

Your sincerely

Bijay Kumar Chaudhary

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ABSTRACT

This report documents the 10-week internship completed at Sajilo Digital Pvt. Ltd. as a Frontend Developer, focusing on the complete modernization and redesign of the Code for Change Nepal website. The primary objective was to migrate an existing static HTML-based website to a dynamic, component-based ReactJS application with improved user experience, performance, and maintainability. The project involved analyzing the existing HTML structure, designing a new component architecture, implementing a responsive design system, and developing interactive features using ReactJS. Key technical implementations included creating reusable React components, implementing client-side routing with React Router, managing application state with React Hooks and Context API, optimizing performance through code splitting and lazy loading, and ensuring full responsiveness across all device types. The redesigned website features a modern user interface, improved navigation structure, dynamic content rendering, interactive elements, and enhanced performance metrics. The migration from static HTML to ReactJS provided the organization with a more maintainable codebase, better development workflow, and foundation for future feature additions. This internship provided comprehensive exposure to the process of modernizing legacy web applications, component-based architecture design, responsive web development, and performance optimization techniques using modern frontend technologies.

Keywords: *ReactJS, Frontend Development, Website Modernization, Component Architecture, Responsive Design, Performance Optimization, Legacy Migration, Web Redesign*

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LIST OF ABBRIVIATION

API – Application Programming Interface

ARIA - Accessible Rich Internet Applications

CMS – Content Management System

CSS – Cascading Style Sheet

ERP – Enterprise Resource Planning

HTML – Hypertext Markup Language

PWA – Progressive Web Apps

QA – Quality Assurance

SEO – Search Engine Optimization

SPA – Single Page Application

SSL – Secure Socket Layer

UI – User Interface

UX - User Experience

WCAG – Web Content Accessibility Guidelines

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CHAPTER 1: INTRODUCTION

1.1. Introduction

In the rapidly evolving digital landscape, non-profit organizations like Code for Change Nepal require modern, engaging, and performant websites to effectively communicate their mission, showcase their work, and engage with volunteers and donors. This internship report details my work as a Frontend Developer at Sajilo Digital Pvt. Ltd., where I undertook the complete redesign and modernization of the Code for Change Nepal website from a static HTML-based platform to a dynamic ReactJS application.

The original website, built with traditional HTML, CSS, and JavaScript, faced limitations in maintainability, user engagement, and performance. The migration to ReactJS provided an opportunity to implement modern web development practices, including component-based architecture, efficient state management, and optimized performance. The redesigned platform aims to better represent the organization's mission of social change through technology while providing an improved user experience for visitors, volunteers, and donors.

This project provided practical experience in analyzing legacy codebases, designing modern architectures, and implementing full website redesigns using contemporary frontend technologies.

1.2. Problem Statement

The existing Code for Change Nepal website presented several challenges that necessitated a complete redesign:

- Outdated Technology Stack: The website was built with static HTML, making content updates difficult and requiring technical expertise for even minor changes.
- Poor Maintainability: Lack of component reusability led to code duplication and inconsistent implementations across pages.
- Limited Interactivity: Minimal JavaScript functionality resulted in a static user experience with limited engagement opportunities.
- Non-Responsive Design: The website did not adapt well to mobile devices, alienating a significant portion of potential visitors.
- Slow Performance: Unoptimized assets and inefficient code structure resulted in slow loading times and poor user experience.

- Inconsistent Design: Visual inconsistencies across pages undermined the organization's professional image.
- Difficult Content Management: Adding new projects, events, or updates required manual HTML editing, limiting the organization's ability to keep content current.
- Poor SEO Structure: The static nature limited opportunities for search engine optimization and content discovery.

The project aimed to redesign the Code for Change Nepal website using modern technologies and best practices. The new website is responsive, interactive, and maintainable, with reusable components, optimized performance, and easier content management, enhancing both the user experience and the organization’s online presence.

1.3. Objectives

The primary objectives of this frontend redesign project were:

- To analyze the existing HTML-based website and identify areas for improvement in structure, design, and functionality.
- To design and implement a modern, component-based architecture using ReactJS for improved maintainability and scalability.
- To create a responsive design system that provides optimal user experience across all device types and screen sizes.
- To implement interactive features and dynamic content rendering to enhance user engagement.
- To optimize website performance through code splitting, asset optimization, and efficient rendering techniques.
- To improve the visual design and user interface while maintaining alignment with the organization's mission and branding.
- To implement a structured content management approach that allows for easier updates and additions.
- To enhance accessibility features ensuring the website is usable by people with disabilities.
- To implement proper SEO structures and metadata management for improved online visibility.

1.4. Scope and Limitation

Scopes:

- Complete frontend redesign and development using ReactJS.
- Migration of all existing content from HTML to React components.
- Implementation of responsive design for mobile, tablet, and desktop devices.
- Development of reusable component library for consistent UI implementation.
- Integration of interactive features (image galleries, event calendars, donation forms).
- Performance optimization including image optimization, code splitting, and lazy loading.
- Implementation of client-side routing for seamless navigation.
- Basic SEO implementation with proper metadata and semantic HTML.
- Deployment to production hosting environment.

Limitations:

- Backend development for content management system was beyond scope; content remains statically managed.
- User authentication and membership features were not implemented in this phase.
- Advanced features like real-time chat, video streaming, or complex data visualizations were not included.
- The project focused on frontend modernization only; backend API integration was limited to mock data.
- Multilingual support was planned but not implemented in the initial release.
- Advanced analytics integration beyond basic Google Analytics was not included.

1.5. Report Organization

This report is structured into four chapters to provide a logical and comprehensive overview of the internship:

- **Chapter 1: Introduction**
  Presents an overview of the project, including the background of the internship, a clear problem statement, the defined objectives, the technical scope and limitations, and the overall organization of this report.
- **Chapter 2: Organization Details and Literature Review**

Discusses the background of Sajilo Digital Pvt. Ltd., including its organizational hierarchy and core working domains. It also covers the description of the intern department and a review of the foundational technologies and existing literature relevant to modern frontend development.

- **Chapter 3: Internship Activities**
  Serves as the technical core of the report, provides a detailed account of my roles, weekly progress, project descriptions, and specific tasks undertaken.
- **Chapter 4: Conclusion and Learning Outcomes**
  Summarizes the project's achievements and reflects on the professional and technical skills acquired.

Additional sections, including References, and Appendices (containing screenshots, source code snippets), are included at the end of the report to provide supporting documentation and evidence of the work performed.

CHAPTER 2: ORGANIZATION DETAILS AND LITERATURE REVIEW

2.1. Introduction to Organization

Sajilo Digital Pvt. Ltd. is a forward-thinking digital solutions agency based in Butwal, Nepal. The company specializes in providing end-to-end technology services, including custom software development, web and mobile application creation, digital marketing, and IT consultancy. Driven by a mission to empower businesses through innovative and efficient digital tools, Sajilo Digital emphasizes agility, quality, and client-centric development methodologies. The company's commitment to adopting modern tech stacks and fostering a culture of continuous learning made it an ideal environment for an immersive internship experience.

2.2. Organizational Hierarchy

The company employs a flat yet functional hierarchy designed to promote efficiency and collaboration:

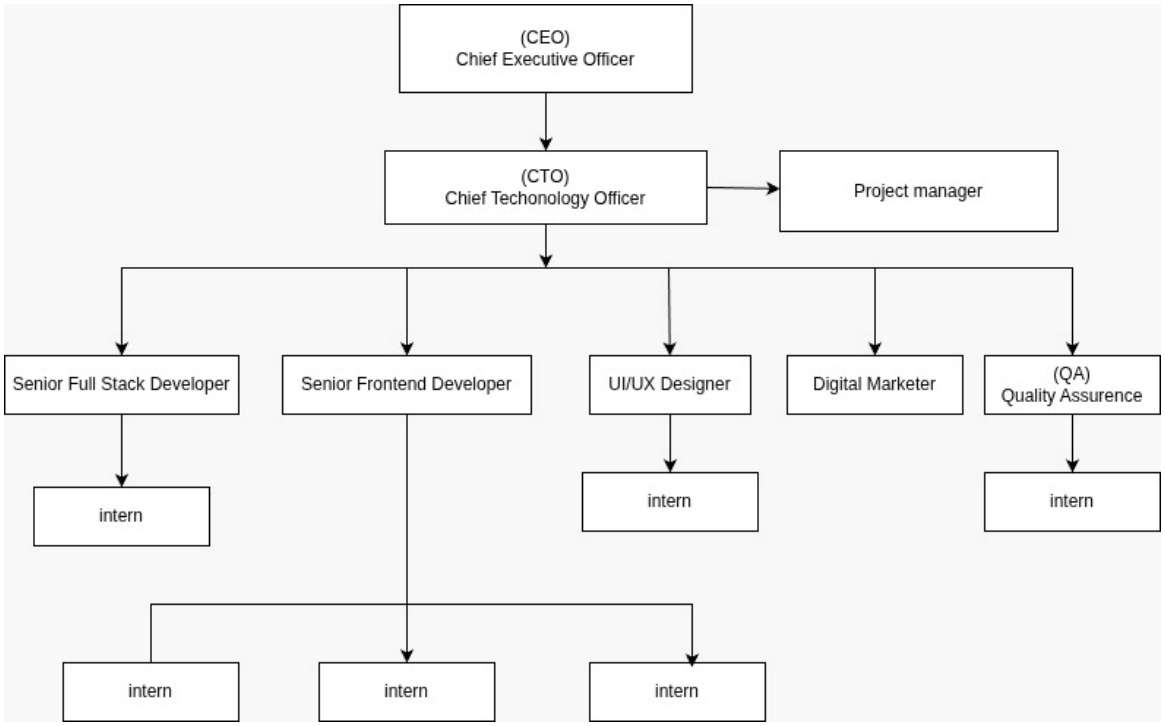

Figure 2.2: Organizational Hierarchy of Sajilo Digital

- Leadership / Directors: Set company vision, strategy, and oversee major client relationships.

- Project Management: Responsible for resource planning, timeline management, and acting as the bridge between clients and the technical team.
- Development Teams: Organized into frontend, backend, and full-stack units, tasked with designing, coding, and testing software products.
- UI/UX Design Team: Creates user interface mock-ups, prototypes, and maintains design systems.
- Quality Assurance (QA) Team: Ensures product reliability through systematic testing procedures.
- Marketing & Sales: Handles business development, branding, and client acquisition.
- Interns: Integrated directly into project teams to contribute to real-world projects under mentorship.

2.3. Working Domains of Organization

Sajilo Digital operates across several key domains within the tech industry:

- Custom Web Development: Building responsive websites, progressive web apps (PWAs), and single-page applications (SPAs) using stacks like Next.js, React, and Node.js.
- Mobile App Development: Creating cross-platform (React Native, Flutter) and native mobile applications.
- Enterprise Software Solutions: Developing tailored business management systems, CRMs, and ERPs.
- E-commerce Solutions: Designing and developing online stores with integrated payment and inventory systems.
- Digital Marketing & SEO: Enhancing online visibility and user engagement through targeted strategies.
- Cloud Services & DevOps: Assisting with deployment, hosting, and infrastructure management.

2.4. Description of the Intern Department / Unit

The internship program at Sajilo Digital is structured to provide maximum practical exposure. Interns are treated as junior developers and assigned to active projects. I was placed in the Frontend Development unit under the guidance of my mentor, Mr. Shailesh Mishra.

The department operated with agile-inspired practices, including daily stand-up meetings, weekly sprint planning, and code review sessions. This environment emphasized collaborative problem-solving, version control best practices using Git, and writing clean, documented code. My primary project was the company website, which gave me direct responsibility over a production-critical asset, from conceptualization to deployment.

Table 2.4: Internship Duration

Start Date:	11/17/2025
Daily Hours:	10:00 AM to 6:00 PM
Working Days:	Sunday to Friday
Position:	Frontend Developer Intern
Company Mentor:	Mr. Sailesh Mishra
College Supervisor:	Mr. Suraj Pandey

2.5. Literature Review / Related Study

The modernization of the Code for Change Nepal website was informed by contemporary frontend development practices:

ReactJS and Component Architecture: Resources like "React Design Patterns and Best Practices" by Carlos Santana Roldán (Roldan, 2019) guided the implementation of reusable, maintainable component structures, emphasizing separation of concerns and single responsibility principle.

Legacy Code Modernization: Research on "Modernizing Legacy Systems" by Robert C. Seacord (Seacord, 2003) informed the approach to analyzing existing HTML structures and developing migration strategies that minimize disruption while maximizing benefits.

Responsive Web Design: Principles from Ethan Marcotte's foundational work (Marcotte, 2011) were applied, utilizing fluid grids, flexible images, and CSS media queries to ensure accessibility across all screen sizes.

Performance Optimization: Studies from Google's Web Vitals initiative and "High Performance Web Sites" by Steve Souders (2007) (Souders, 2007) informed techniques for minimizing load times through asset optimization, code splitting, and efficient rendering.

Accessibility Standards: WCAG 2.1 guidelines and "Inclusive Design Patterns" by Heydon Pickering (Pickering, 2016) influenced implementation decisions to ensure the website is usable by people with diverse abilities.

SEO Best Practices: Current search engine optimization guidelines from Google's Webmaster Central informed implementation of semantic HTML, metadata, and site structure improvements.

CHAPTER 3: INTERNSHIP ACTIVITIES

3.1. Roles and Responsibilities

As a Frontend Developer Intern, my responsibilities focused on the complete redesign and modernization process:

- Legacy Code Analysis: Analyzing the existing HTML-based website structure, identifying reusable patterns, and documenting technical debt.
- Component Architecture Design: Designing a modular component structure for the React application, identifying reusable UI components and their relationships.
- React Component Development: Building reusable React components following atomic design principles (atoms, molecules, organisms, templates, pages).
- Responsive Design Implementation: Creating responsive layouts using CSS Grid, Flexbox, and media queries with a mobile-first approach.
- State Management: Implementing application state management using React Hooks (useState, useEffect) for interactive features.
- Client-Side Routing: Implementing navigation with React Router for seamless single-page application experience.
- Performance Optimization: Optimizing images, implementing code splitting, lazy loading, and other performance improvements.
- Accessibility Implementation: Ensuring the website meets accessibility standards through semantic HTML, ARIA attributes, and keyboard navigation.
- Content Migration: Migrating existing content from HTML files to React components while improving structure and presentation.
- Interactive Features Development: Implementing interactive elements like image galleries, event calendars, donation forms, and animated transitions.
- Testing and Quality Assurance: Conducting cross-browser testing, responsive testing, and accessibility testing.
- Deployment: Configuring and deploying the React application to production hosting environment.
- Documentation: Creating component documentation, style guides, and development guidelines.

3.2. Weekly Log

A detailed breakdown of activities, responsibilities, and key learnings for each week:

Weekly Internship Log

Name: Bijay Kumar Chaudhary

Project: Code for Change Nepal Website Redesign

Week no: 1	**Date: 11/17/2025 - 11/23/2025**

Responsibilities:

- Project initiation and legacy code analysis.

Activities:

- Conducted comprehensive analysis of existing HTML website.
- Documented structure, components, and content.
- Identified pain points and improvement opportunities.
- Set up development environment and React project structure.

Observation:

- Thorough analysis of existing systems reveals hidden patterns and informs better architectural decisions.

Performance Evaluation by Mentor:

- Demonstrated strong analytical skills and initiative in understanding legacy systems and setting up a solid project foundation.

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Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bijay Kumar Chaudhary

Project: Code for Change Nepal Website Redesign

Week no: 2	**Date: 11/24/2025 - 11/30/2025**

Responsibilities:

- Component architecture and design system planning

Activities:

- Designed component hierarchy and file structure.
- Created design tokens (colors, typography, spacing).
- Developed initial style guide.
- Planned migration strategy from HTML to React components.

Observation:

- Investing time in planning component architecture significantly accelerates development and ensures consistency.

Performance Evaluation by Mentor:

- Showed excellent planning ability by designing a scalable component architecture and consistent design system.

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Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bijay Kumar Chaudhary

Project: Code for Change Nepal Website Redesign

Week no: 3	**Date: 12/01/2025 - 12/07/2025**

Responsibilities:

- Foundation components development.

Activities:

- Built foundational layout components (Header, Footer, Navigation, Layout).
- Created base UI components (Button, Card, Section, Container).
- Implemented responsive grid system.
- Set up CSS architecture with CSS Modules.

Observation:

- A robust foundation of reusable components is crucial for efficient development of complex interfaces.

Performance Evaluation by Mentor:

- Effectively developed reusable and well-structured foundational components, ensuring a robust frontend base.

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Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bijay Kumar Chaudhary

Project: Code for Change Nepal Website Redesign

Week no: 4	**Date: 12/08/2025 - 12/14/2025**

Responsibilities:

- Page components and routing implementation.

Activities:

- Developed main page components (Home, About, Provinces, Events, Blog, Our Impact etc).
- Implemented client-side routing with React Router.
- Created page templates and layouts.
- Began content migration from HTML to React components.

Observation:

- Component-based architecture makes content migration systematic and manageable.

Performance Evaluation by Mentor:

- Successfully implemented core pages and routing with a systematic approach to content migration.

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Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bijay Kumar Chaudhary

Project: Code for Change Nepal Website Redesign

Week no: 5	**Date: 12/15/2025 - 12/21/2025**

Responsibilities:

- Interactive features and state management.

Activities:

- Implemented interactive image galleries for project showcases.
- Added provinces section with filtering.
- Created donation form with validation.
- Implemented state management for interactive elements using React Hooks.

Observation:

- Interactive features significantly enhance user engagement and convey professionalism.

Performance Evaluation by Mentor:

- Displayed good problem-solving skills by integrating interactive features and managing application state efficiently.

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Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bijay Kumar Chaudhary

Project: Code for Change Nepal Website Redesign

Week no: 6	**Date: 12/22/2025 - 12/28/2025**

Responsibilities:

- Responsive design refinement.

Activities:

- Conducted comprehensive responsive testing across devices.
- Refined breakpoints and adaptive layouts.
- Optimized touch interactions for mobile devices.
- Fixed responsive design issues and inconsistencies.

Observation:

- Mobile responsiveness is non-negotiable in modern web development and requires continuous testing.

Performance Evaluation by Mentor:

- Exhibited attention to detail by refining responsiveness and ensuring a seamless cross-device user experience.

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Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bijay Kumar Chaudhary

Project: Code for Change Nepal Website Redesign

Week no: 7	**Date: 12/29/2025 - 01/04/2026**

Responsibilities:

- Performance optimization.

Activities:

- Optimized all images using modern formats and compression.
- Implemented code splitting and lazy loading for routes.
- Reduced JavaScript bundle size.
- Added loading states and performance monitoring.

Observation:

- Performance optimization has a direct impact on user retention and search engine rankings.

Performance Evaluation by Mentor:

- Demonstrated strong optimization skills by improving performance and reducing load times effectively.

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Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bijay Kumar Chaudhary

Project: Code for Change Nepal Website Redesign

Week no: 8	**Date: 01/05/2026 - 01/11/2026**

Responsibilities:

- Accessibility and Testing.

Activities:

- Conducted cross-browser testing (Chrome, Firefox, Safari, Edge).
- Implemented semantic HTML structure throughout.
- Tested form validations and other UI components

Observation:

- Accessibility and SEO are foundational aspects that should be integrated from the beginning.

Performance Evaluation by Mentor:

- Maintained quality standards through thorough testing and implementation of accessibility best practices.

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Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bijay Kumar Chaudhary

Project: Code for Change Nepal Website Redesign

Week no: 9	**Date: 01/12/2026 - 01/18/2026**

Responsibilities:

- Final content migration and polishing.

Activities:

- Completed migration of all remaining content.
- Applied final styling polish and animations.
- Conducted user acceptance testing with stakeholders.
- Fixed remaining bugs and issues.

Observation:

- Attention to detail in the final stages significantly elevates perceived quality.

Performance Evaluation by Mentor:

- Showed dedication and precision in finalizing content, fixing bugs, and enhancing overall UI/UX quality.

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Mentor's Approval

Mr. Shailesh Mishra

Weekly Internship Log

Name: Bijay Kumar Chaudhary

Project: Code for Change Nepal Website Redesign

Week no: 10	**Date: 01/19/2026 - 01/23/2026**

Responsibilities:

- Deployment and documentation.

Activities:

- Deployed React application to production hosting.
- Configured custom domain and SSL.
- Created comprehensive documentation for components and styles.
- Prepared project handover materials and final report.

Observation:

- Proper deployment and documentation ensure the sustainability and maintainability of the project.

Performance Evaluation by Mentor:

- Successfully handled deployment and documentation, ensuring project readiness and long-term maintainability.

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Mentor's Approval

Mr. Shailesh Mishra

3.3. Description of the Project (s) Involved During Internship

Primary Project: Code for Change Nepal Website Redesign (Next.js)

A complete modernization project migrating a static HTML website to a dynamic ReactJS application for a non-profit organization.

- **Live URL:** Code for Change
- **Technical Specification:**
  i. Frontend Framework: React 18 with functional components and hooks
  ii. Routing: React Router v6 for client-side navigation
  iii. State Management: React Hooks (useState, useEffect, useContext) for local and global state
  iv. Styling: CSS Modules for component-scoped styles with custom design system
  v. Build Tool: Create React App with custom configuration
  vi. Performance: Code splitting, lazy loading, image optimization
  vii. Deployment: Vercel hosting platform
- **Key Features and Implementation:**
  i. Component-Based Architecture: 40+ reusable React components organized by atomic design principles
  ii. Responsive Design System: Fully responsive layouts with mobile-first approach
  iii. Interactive Elements: Image galleries, internship form, donation forms with validation
  iv. Performance Optimized: 85+ Lighthouse performance score with optimized assets
  v. Accessibility Compliant: WCAG 2.1 AA compliance with semantic HTML and ARIA attributes
  vi. SEO Friendly: Proper metadata, structured data, and semantic markup - Modern Design: Clean, professional interface aligning with non-profit branding
  vii. Easy Maintenance: Modular structure enabling easy updates and additions.

Auxiliary Project 1: Medical App – Doctor Appointment

A frontend project focused on dynamic user interfaces for healthcare services.

- **Objective:** Build a responsive web application for patients to book doctor appointments, view schedules, and manage consultations.
- **Tech:** Next.js, Tailwind CSS
- **Learning Outcomes:** Enhanced skills in state management, form handling, conditional rendering, responsive design.

- **Repository:** https://github.com/sajilodigital9000/python-server

Auxiliary Project 2: HRM of Sajilo Digital

A frontend-focused project to develop employee management dashboards.

- **Objective:** Create an interactive Human Resource Management interface to manage employee records, attendance.
- **Tech:** React, React Router, Tailwind CSS, React icons.
- **Learning Outcomes:** Improved skills in designing modular UI components, data tables, interactive dashboards, and frontend validation for HR workflows.

3.4. Tasks / Activities Performed

This section provides a detailed breakdown of the technical and professional activities carried out during the 10-week internship. Each activity is elaborated with the context, approach, challenges encountered, and solutions implemented.

3.4.1 Project Setup and Environment Configuration

At the outset of the internship, I established a robust development environment for the Sajilo Digital website. This involved:

Selecting the Tech Stack: After evaluating options, we chose Next.js 14 with React 18 for its hybrid rendering capabilities (SSR/SSG), built-in performance optimizations, and excellent developer experience. Vercel was selected for deployment due to its seamless integration with Next.js.

Repository Initialization: I created a Git repository on GitHub, setting up a branching strategy with main for production, develop for integration, and feature branches for each new component or page. This ensured clean version control and facilitated team collaboration.

Tooling Configuration: I configured ESLint and Prettier to enforce consistent code style and catch potential errors early. Custom rules were added to align with the company's coding standards.

Environment Variables: I set up .env.local for development secrets (e.g., API keys, contact form endpoints) and documented the required variables in a .env.example file.

Development Scripts: I enhanced the package.json scripts with commands for running the dev server, building for production, and running lint checks. I also integrated next-optimized-images to handle image optimization during builds.

Challenges and Solutions: Initially, there were conflicts between ESLint and Prettier that caused inconsistent formatting. I resolved this by installing the eslint-config-prettier package and adjusting the .eslintrc configuration to defer stylistic rules to Prettier.

3.4.2 Component Architecture and Design System Implementation

To ensure maintainability and consistency, I designed a modular component architecture following the atomic design principle:

Atoms: Created basic building blocks such as Button, Input, Icon, Heading, Text, and Container. These components received props for customization (size, variant, color) and included proper ARIA labels for accessibility.

Molecules: Combined atoms to form functional groups like NavLink, SearchBar, Card, ServiceCard, and TestimonialCard. For example, the ServiceCard accepts an icon, title, description, and a link, rendering a consistent card layout throughout the services section.

Organisms: Developed larger sections that combine molecules, such as HeroSection, FeaturesGrid, ProjectShowcase, and ContactForm. These organisms handled data fetching (via props) and layout logic.

Templates and Pages: Created page templates (e.g., HomeTemplate, ServicesTemplate) that arranged organisms into a page layout. Each Next.js page imported the corresponding template and passed in the required data.

Documentation: I documented each component using JSDoc comments, specifying props, default values, and usage examples. This made it easier for other team members to understand and reuse components.

Challenges and Solutions: One challenge was ensuring that components remained truly reusable without becoming overly complex. I solved this by applying the single responsibility principle—each component does one thing well—and by using composition over configuration. For instance, the Button component accepts an as prop to render as a <button> or <a> element, maintaining flexibility.

3.4.3 Responsive Layout Development

A key requirement was a fully responsive design that works flawlessly on desktops, tablets, and mobile devices. I adopted a mobile-first approach:

CSS Methodology: I used CSS Modules to scope styles to individual components, preventing global conflicts. A global globals.css defined CSS variables for colors, typography, spacing, and breakpoints.

Breakpoints: Based on common device sizes, I defined breakpoints: 480px (mobile), 768px (tablet), 1024px (desktop), and 1440px (large desktop). Media queries were used to adjust layouts, font sizes, and spacing accordingly.

Layout Components: I created layout components like Container, Grid, and Flex that encapsulate responsive behaviors. The Grid component, for example, uses CSS Grid and accepts props for columns, gaps, and responsiveness (e.g., gridTemplateColumns={{ base: '1fr', md: '1fr 1fr', lg: '1fr 1fr 1fr' }}).

Navigation: The main navigation was implemented as a Navbar component that collapses into a hamburger menu on mobile. I used useState to control the mobile menu's open/close state and added a smooth transition using CSS transforms.

Testing: I tested the responsiveness using browser dev tools' device emulation and physical devices (iPhone, iPad, various Android phones). I made extensive use of Chrome DevTools' responsive design mode to iterate quickly.

Challenges and Solutions: Handling complex layouts like the portfolio grid that needed to adapt from 3 columns on desktop to 2 on tablet to 1 on mobile required careful planning. I solved this by using CSS Grid with repeat(auto-fit, minmax(280px, 1fr)) combined with media queries to fine-tune the column count for optimal image sizing.

3.4.4 Data Fetching and State Management

Leveraging Next.js data fetching capabilities was central to the project:

Static Generation (SSG): For blog posts and portfolio items, I used getStaticProps to fetch content from local markdown files at build time. I implemented a utility to parse frontmatter and content using gray-matter. This produced static HTML files that load extremely fast.

Server-Side Rendering (SSR): For pages requiring real-time data (e.g., a live service status or dynamic content), I used getServerSideProps. This was applied to the contact page to pre-populate a location map based on IP geolocation (a third-party API).

Client-Side State: For interactive elements like the contact form, I used React hooks (useState, useEffect) to manage form state, validation, and submission status. I also used useContext to share a theme preference across the site (light/dark mode) without prop-drilling.

API Integration: I set up API routes inside Next.js (/api/contact) to handle form submissions and forward them to a backend service. This kept sensitive keys secure and allowed for server-side validation.

Challenges and Solutions: One challenge was ensuring that dynamic pages (like blog posts) were correctly routed and displayed. Next.js dynamic routing with [slug].js was used, but we needed to generate a list of all slugs at build time. I implemented getStaticPaths to return all possible slugs, and getStaticProps to fetch the content for a specific slug.

3.4.5 Performance Optimization

Performance was a top priority to achieve high Lighthouse scores and a fast user experience:

Image Optimization: All images were served via Next.js's <Image> component, which automatically optimizes formats (WebP), resizes, and lazy loads. I set appropriate sizes attributes to serve responsive images.

Code Splitting and Lazy Loading: Next.js automatically splits code by page, but I also used dynamic imports (next/dynamic) for components that are not needed immediately, such as the contact form and heavy animations. I added loading skeletons to improve perceived performance.

Bundle Analysis: I used @next/bundle-analyzer to inspect the JavaScript bundle and identified that a large charting library was unnecessarily included. I replaced it with a lightweight alternative.

Font Optimization: I hosted custom fonts locally using next/font to reduce external requests and improve privacy. I also used font-display: swap to avoid layout shifts.

Caching Headers: I configured caching headers in the next.config.js for static assets (e.g., images, fonts) to leverage browser caching.

Results: After optimization, Lighthouse scores consistently exceeded 90 for performance, accessibility, best practices, and SEO.

3.4.6 Cross-Browser and Accessibility Testing

To ensure a consistent experience across all major browsers and devices, I performed extensive testing:

Browser Testing: I manually tested on Chrome, Firefox, Safari, and Edge on Windows and macOS. I used BrowserStack to test on older browser versions.

Accessibility Testing: I used the WAVE tool and Lighthouse accessibility audits to identify issues. I then implemented fixes such as adding alt attributes to images, ensuring proper heading hierarchy, and adding aria-label to interactive elements.

Keyboard Navigation: I tested the entire site using only keyboard navigation, ensuring that all interactive elements are focusable and that focus order is logical. I added :focus-visible styles for better visual feedback.

Challenges and Solutions: Safari had a specific issue with CSS Grid layout for the portfolio grid. I discovered that Safari required -webkit- prefixes for certain grid properties. I used PostCSS with Autoprefixer to automatically add prefixes during the build process.

3.4.7 Deployment and Continuous Integration

The final phase involved deploying the website to production:

Vercel Setup: I connected the GitHub repository to Vercel, configuring automatic deployments for the main branch. I set up preview deployments for pull requests, allowing the team to review changes before merging.

Custom Domain and SSL: I configured the custom domain sajilodigital.com.np through Vercel's domain management, which automatically provisions an SSL certificate.

Environment Variables: I added all required environment variables to Vercel's project settings, ensuring they were available in the production environment.

Monitoring: I enabled Vercel Analytics to track performance metrics and error logging to quickly identify any issues after launch.

3.4.8 Documentation and Knowledge Transfer

Throughout the internship, I maintained comprehensive documentation:

Component Documentation: I created a Storybook instance for the component library, documenting each component's props, usage, and examples.

Project Setup Guide: I wrote a README.md with detailed instructions on setting up the project, configuring environment variables, and running the development server.

Deployment Notes: I documented the deployment process, including Vercel configuration, environment variables, and post-deployment verification steps.

Code Comments: I ensured that all complex logic was accompanied by clear comments explaining the reasoning behind the implementation.

3.4.9 Collaboration and Communication

I actively participated in daily stand-up meetings, where I reported progress, highlighted blockers, and coordinated with the mentor. I used Slack for real-time communication and GitHub issues to track tasks and bugs. During code reviews, I provided constructive feedback on peers' pull requests and incorporated feedback into my own work. This collaborative environment taught me the importance of clear communication and teamwork in software development.

In summary, the 10-week internship involved a wide range of activities that collectively contributed to the successful delivery of a production-grade website and significantly enhanced my technical and professional skills.

CHAPTER 4: CONCLUSION AND LEARNING OUTCOMES

4.1. Conclusion

The 10-week internship at Sajilo Digital Pvt. Ltd. provided an immersive experience in modern frontend development through the complete redesign and modernization of the Code for Change Nepal website. This project successfully transformed a static HTML-based website into a dynamic, performant ReactJS application, demonstrating practical application of modern web development principles to solve real-world problems.

The migration from legacy HTML to ReactJS provided the organization with a more maintainable, scalable, and engaging digital presence while improving technical metrics across performance, accessibility, and SEO. The experience of analyzing existing systems, designing modern architectures, and implementing complete redesigns strengthened my understanding of the software modernization process and component-based development.

This internship effectively bridged theoretical knowledge with practical implementation, providing clarity on architectural decisions, performance considerations, and user experience principles that are critical in professional web development.

4.2. Learning Outcomes

Technical Skills:

- ReactJS Mastery: Gained extensive experience with React hooks, component lifecycle, context API, and modern React patterns.
- Legacy System Analysis: Developed skills in analyzing existing codebases, identifying improvement opportunities, and planning migration strategies.
- Component Architecture Design: Learned to design and implement scalable component architectures following atomic design principles.
- Responsive Web Development: Mastered responsive design techniques using CSS Grid, Flexbox, and media queries with mobile-first approach.
- Performance Optimization: Applied techniques for improving Core Web Vitals including image optimization, code splitting, and bundle size reduction.
- Accessibility Implementation: Gained practical experience implementing WCAG 2.1 standards with semantic HTML

- Deployment and DevOps: Gained experience with modern deployment workflows and hosting platforms.

Professional Skills:

- Project Planning: Developed ability to plan and execute complex migration projects with multiple phases.
- Problem-Solving: Enhanced debugging and problem-solving skills across the complete modernization process.
- Client Communication: Improved ability to understand client needs and translate them into technical requirements.
- Documentation: Created comprehensive technical documentation for components, styles, and deployment procedures.
- Quality Assurance: Developed systematic testing approaches for responsive design, accessibility, and cross-browser compatibility.
- Time Management: Learned to manage complex projects with multiple components and deadlines effectively.

The internship has equipped me with both the technical depth and professional mindset required for a career in frontend web development, providing confidence to tackle complex web projects in professional settings.

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APPENDICES

Screenshots

i. Home page

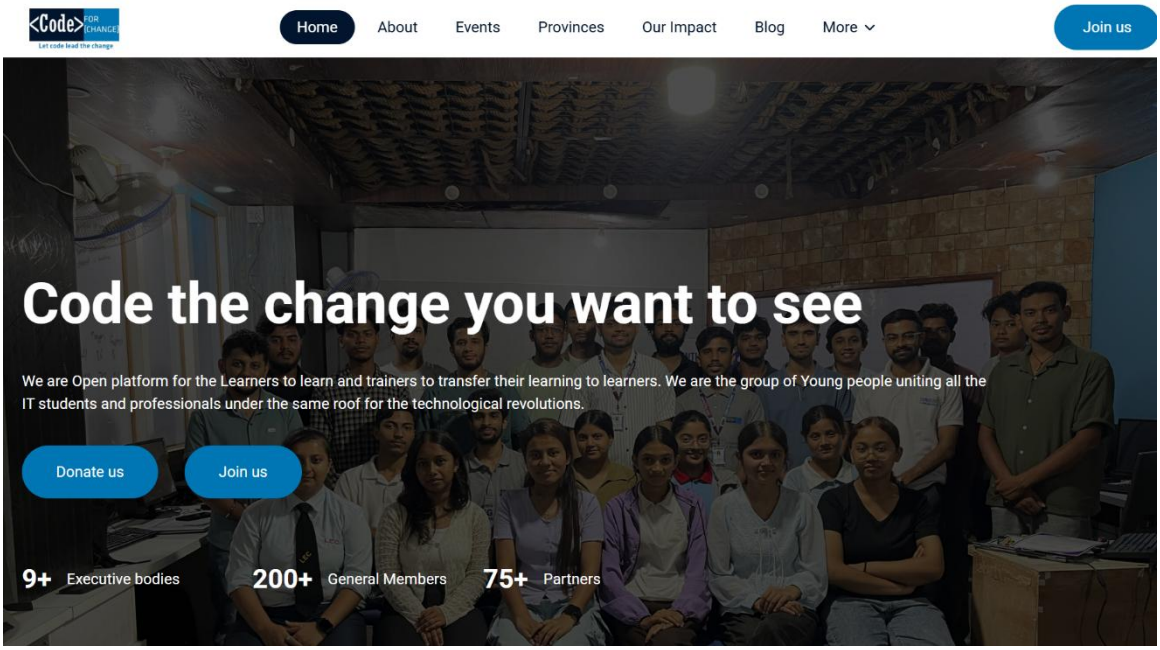

Figure i: Home page of Code for change

ii. About page

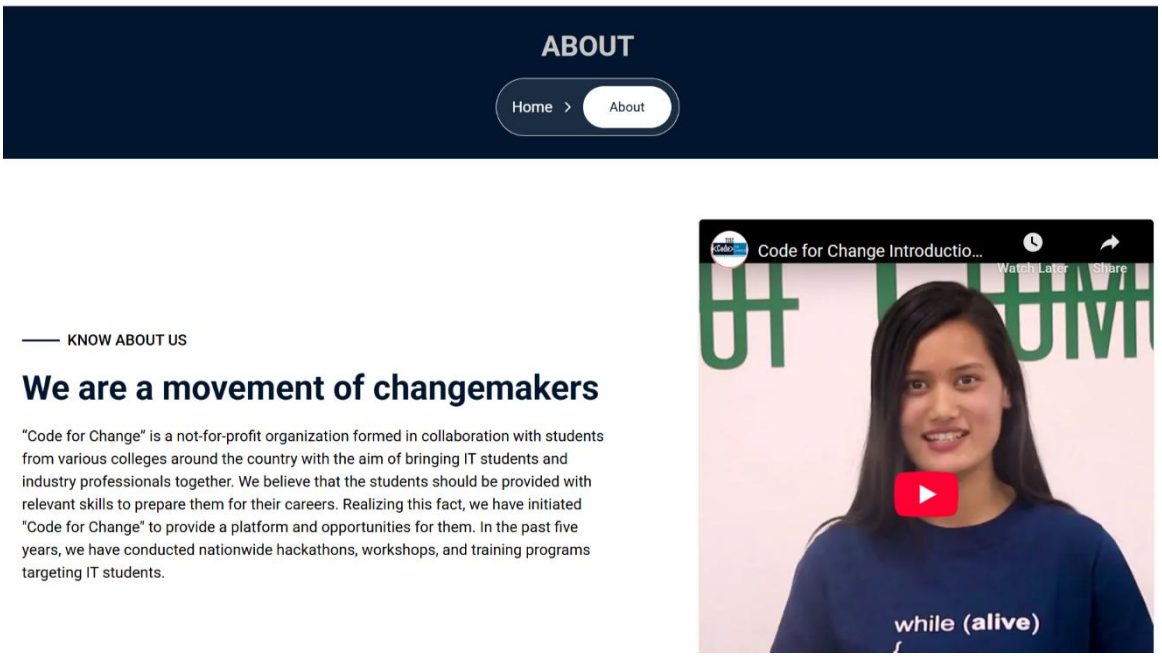

Figure ii: About page of Code for change

iii. Blog page

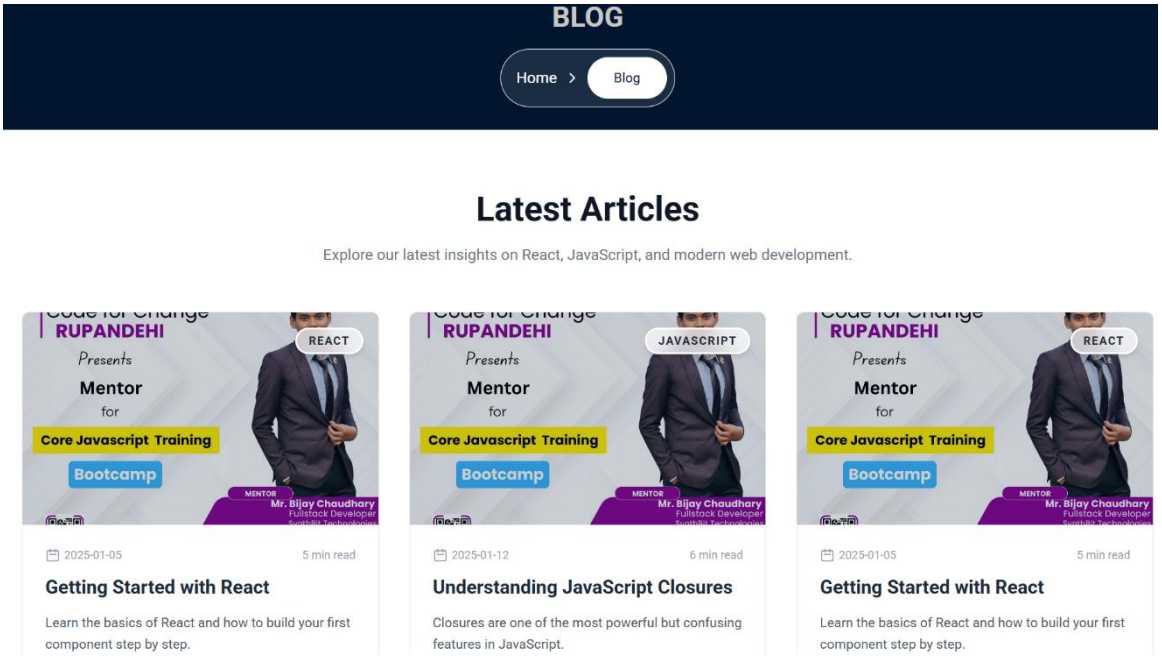

Figure iii: Blog page of Code for change

iv. Event page

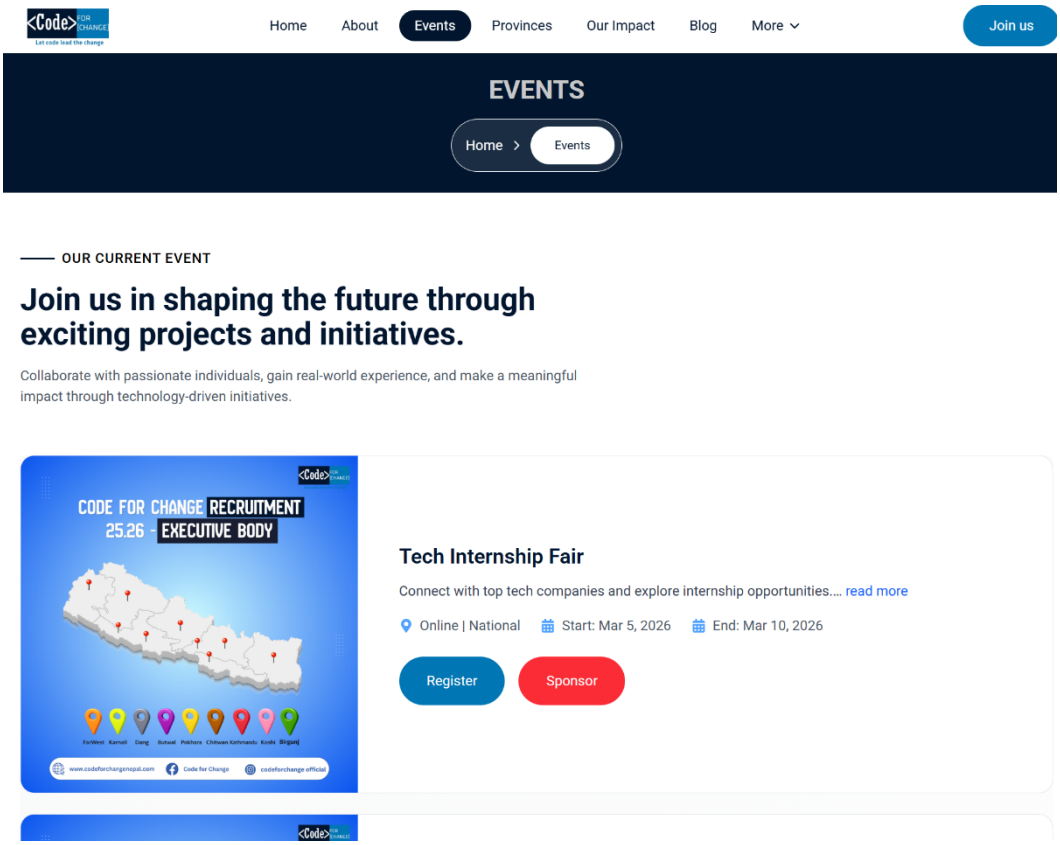

Figure iv: Event page of Code for change

v. Gallery page

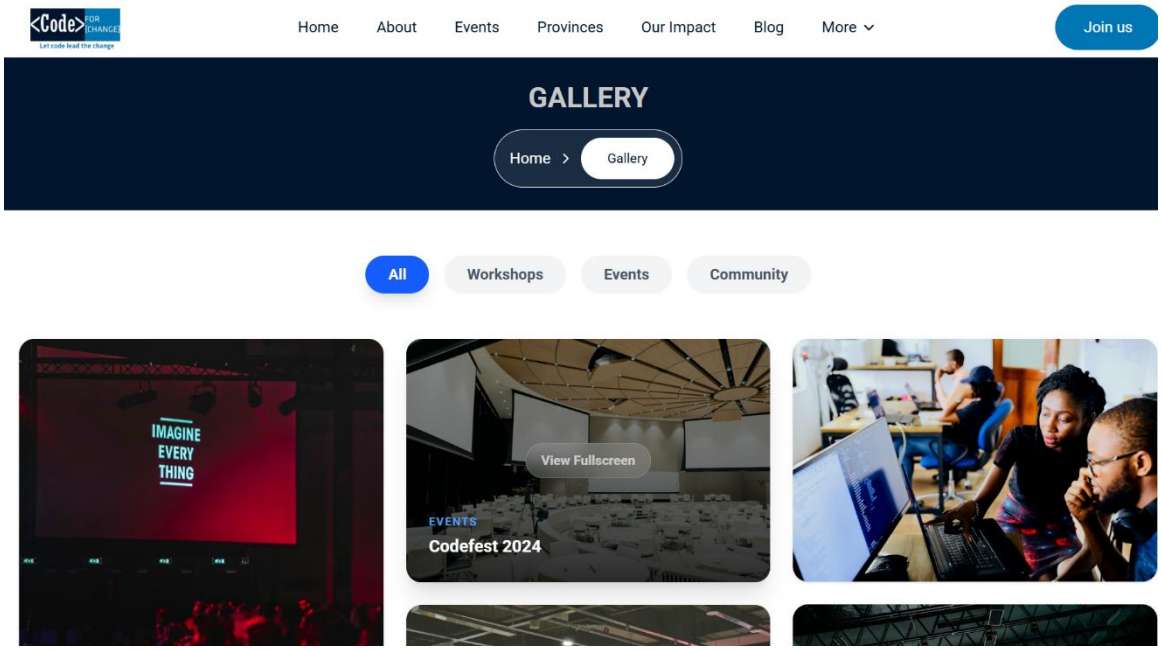

Figure v: Gallery page of Code for change

vi. Our Impact page

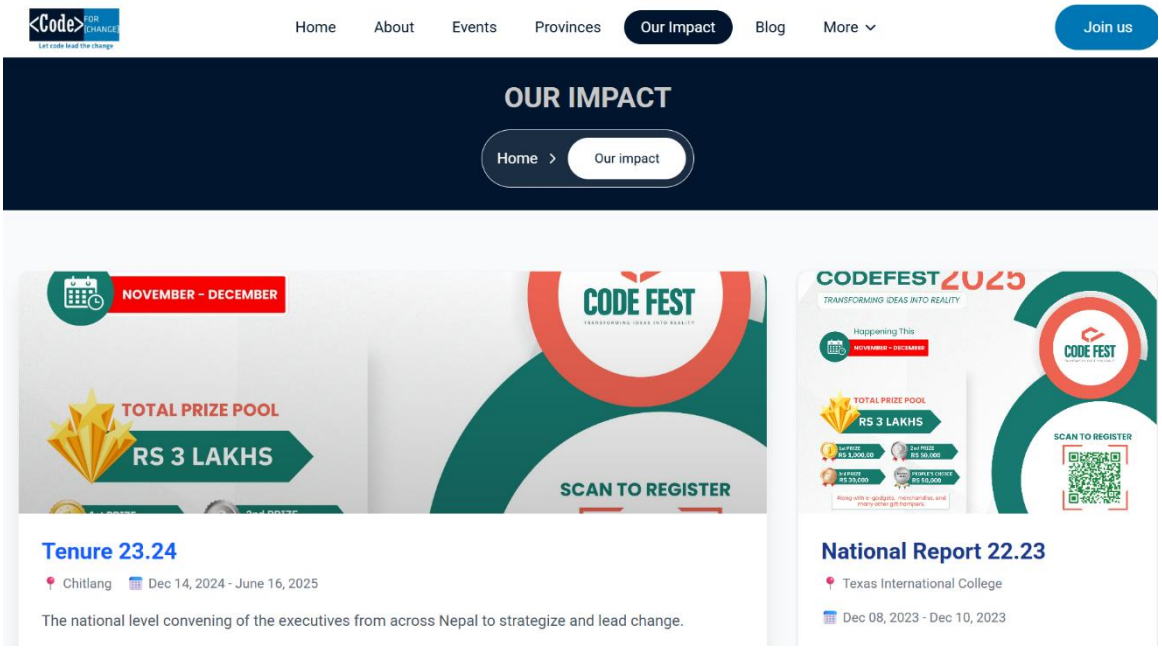

Figure vi: Our Impact page of Code for change

vii. Province page

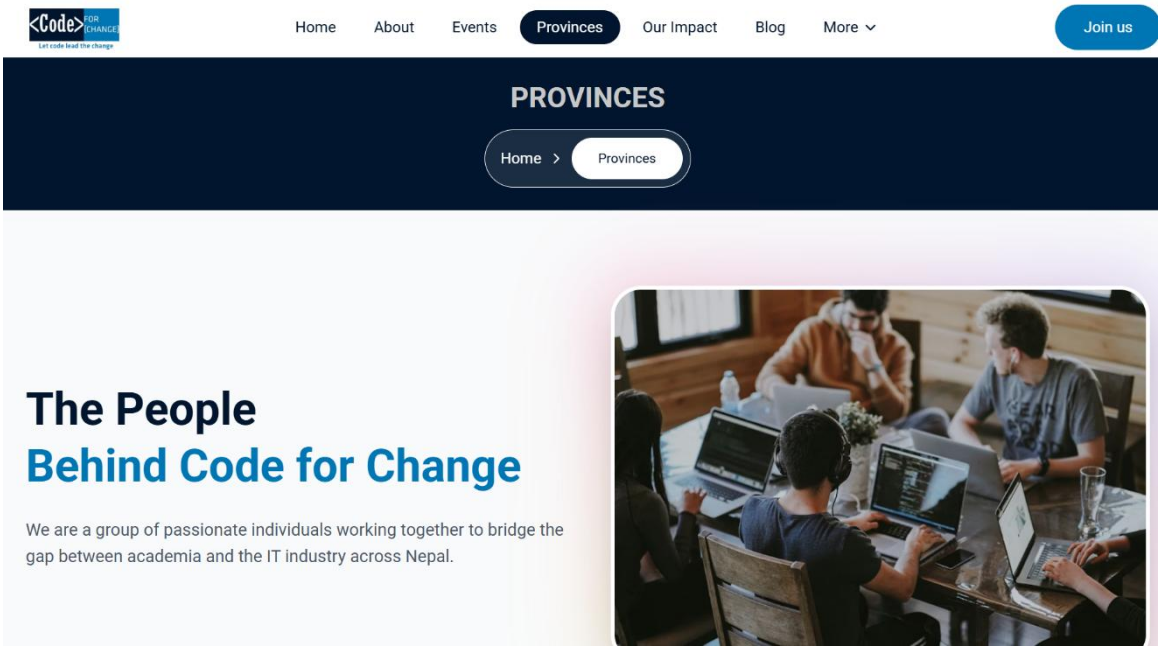

Figure vii: Province page of Code for change

viii. Register page

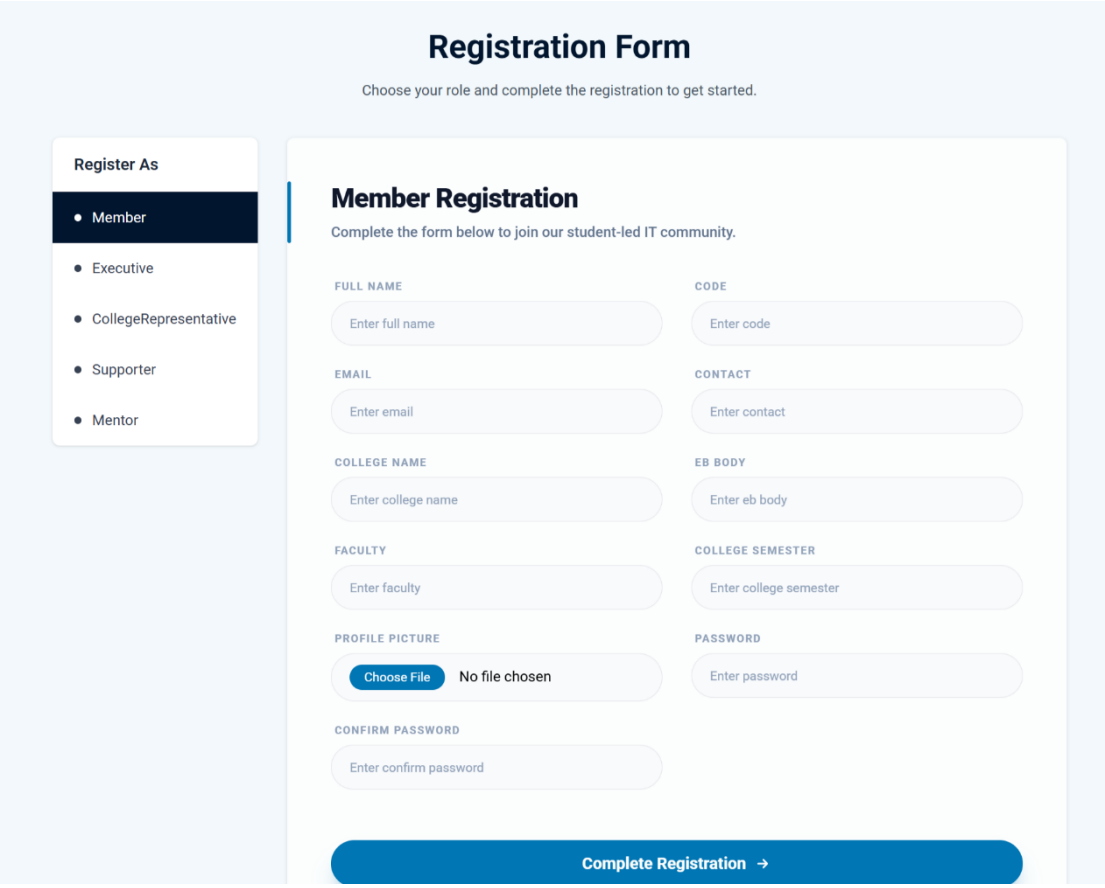

Figure viii: Register page of Code for change

ix. Contact page

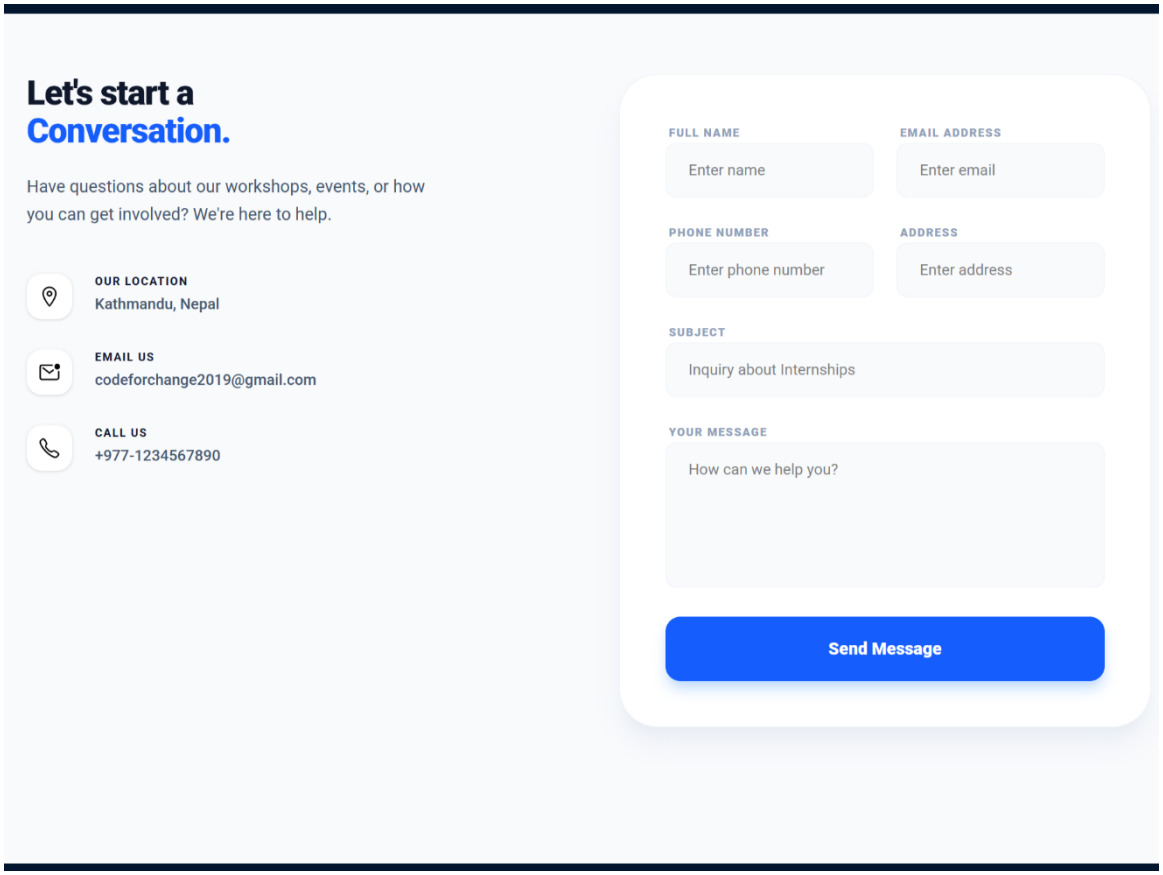

Figure ix: Contact page of Code for change